

6) Inhomogeneous Solution: Dipole Radiation

① A general model of radiation

In the previous section, we examined solutions to the homogeneous wave equation, which describes wave propagation in source-free regions. To understand how electromagnetic waves are generated by charge and current distributions, we must now turn to the inhomogeneous wave equation, where source terms play a central role.

This section focuses on one of the most fundamental radiation scenarios: dipole radiation from a harmonically oscillating electric dipole. Despite the conceptual simplicity of a dipole system, we will see that its radiative solutions are complex.

A key physical insight guides our analysis: radiated fields must propagate outward from the source, with no incoming waves from infinity (a causal boundary condition). Consequently, we seek only the particular solution to the inhomogeneous wave equation – one that corresponds purely to outgoing radiation fields.

Before delving into the mathematical formalism, let us first examine an intuitive example that reveals profound physical insights about radiation. The model is illustrated in the figure below.

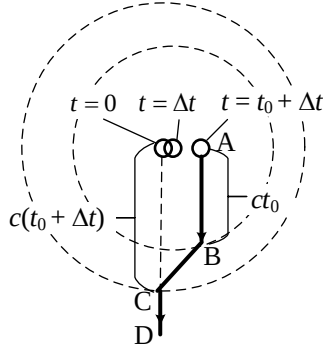


Figure. A moving charge, and one of its representative electric field line

A point charge is initially at rest. At $t = 0$, it begins accelerating at a constant rate a along the $+x$ direction for a short duration Δt . Its position is denoted as x , so $\ddot{x} = a$. After this acceleration phase, the charge continues moving at a constant velocity $\dot{x} = v = a\Delta t$. At $\Delta t + t_0$ (having moved at a constant speed for a period t_0), we take a snapshot of the field. In the figure, the charge's positions at $t = 0$, $t = \Delta t$, and $t = \Delta t + t_0$ are marked by small solid circles.

The information about the charge's movement propagates outward as a spherical wave. The larger (outer) dashed circle represents the region within which the message that the charge is no longer stationary has been received. At time $t = \Delta t + t_0$, this sphere has a radius of $c(\Delta t + t_0)$, where c

is the speed of light. Meanwhile, the message that the charge has begun moving at a constant velocity has reached only a smaller spherical region with a radius of ct_0 , represented by the inner dashed circle.

Inside the inner sphere, all points have been "notified" that the charge is now moving at a constant velocity. As a result, the field line AB is straight, characteristic of a uniformly moving charge. Outside the outer sphere, no information about the charge's motion has yet arrived. Thus, the field line CD remains straight, corresponding to the unmodified field of the original static charge. However, in the transition region between the two spheres, the field is not yet fully updated. Here, the field line develops a kink (section BC), connecting the inner updated field to the outer static field. Consequently, the overall field line (highlighted in bold) is no longer straight but exhibits a propagating distortion.

This kink moves outward at the speed of light, carrying energy with it. Crucially, this transient, non-uniform part of the field is responsible for electromagnetic radiation. The figure below shows a magnified view of the kink in the field line. Let $E_{\parallel}$ denote the field component parallel to the direction of light propagation and $E_{\perp}$ the perpendicular component.

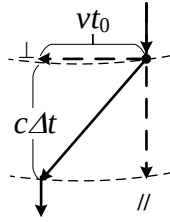


Figure. Zoomed-in view of the field kink.

The geometry shown in the figure suggests the following relationship:

$$\frac{E_{\parallel}}{E_{\perp}} = \frac{c\Delta t}{vt_0} = \frac{c\Delta t}{(a\Delta t)t_0} = \frac{c}{at_0}.$$

From this, we deduce:

$$E_{\perp} = \frac{at_0}{c} E_{\parallel}.$$

The parallel component can be estimated using Coulomb's Law:

$$E_{\parallel} = \frac{q}{4\pi\epsilon_0 r^2}.$$

Since $r \approx ct_0$, we can substitute $t_0 \approx r/c$ into the expression for $E_{\perp}$, thus:

$$E_{\perp} = \frac{ar}{c^2} \cdot \frac{q}{4\pi\epsilon_0 r^2} = \frac{q}{4\pi\epsilon_0 c^2} \frac{a}{r}.$$

Note that this analysis is performed along the direction perpendicular to the particle's motion.

Along the particle's trajectory, we observe that there is no kink, and consequently, $E_{\perp} = 0$.

Therefore, we expect a dependence on $\sin \theta$, and the final expression is:

$$E_{\perp} \approx \frac{q}{4\pi\epsilon_0 c^2} \frac{a}{r} \sin \theta.$$

The analysis reveals several key properties of the radiated field:

- (a) The field decays as $1/r$, rather than $1/r^2$.
- (b) It is proportional to the acceleration of the charge; charges moving at constant velocity do not generate radiation.
- (c) The radiated electric field has a component perpendicular to the direction of wave propagation.
- (d) There is an angular dependence: the radiated energy is maximized in directions perpendicular to the acceleration and vanishes along the direction of acceleration.

② Retarded potentials

Having grasped the basic concept of radiation, let us now delve into a deeper exploration of the underlying physics. In the case of dynamic charge and current distributions (i.e., "sources"), the inhomogeneous wave equations govern the behavior:

$$\begin{cases} \square^2 \phi = -\rho/\epsilon_0 \\ \square^2 \vec{A} = -\mu_0 \vec{J} \end{cases}.$$

These equations can be solved using Green's function. Taking the equation for ϕ as an example,

$$\square^2 \phi = \left(\nabla^2 - \frac{1}{c^2} \partial_{tt} \right) \phi(\vec{r}, t) = -\frac{\rho(\vec{r}, t)}{\epsilon_0},$$

the idea behind the method of Green's function is to first find the solution corresponding to a "point" source, and then apply the superposition principle to construct the final solution by convolving with the spatiotemporal distribution of the source. A spatiotemporal point source (i.e., an impulse) is represented by the Dirac delta function $\delta(\vec{r} - \vec{r}')\delta(t - t')$, meaning that the source is localized at position $\vec{r}'$ and time t' . The solution to this point source is called the Green's function, denoted by $G(\vec{r}, t; \vec{r}', t')$, which gives the field at location $\vec{r}$ and time t in response to a source at position $\vec{r}'$ and time t' :

$$\left(\nabla^2 - \frac{1}{c^2} \partial_{tt} \right) \underbrace{G(\vec{r}, t; \vec{r}', t')}_{\text{response at } \vec{r}, t} = - \underbrace{\delta(\vec{r} - \vec{r}')\delta(t - t')}_{\text{Impulse at } \vec{r}', t'}.$$

To solve the above equation, we employ the technique of Fourier transformation. The Fourier transform of the Green's function is expressed as:

$$g(\vec{k}, \omega) = \iint G(\vec{r}, t; \vec{r}', t') e^{-j\vec{k} \cdot (\vec{r} - \vec{r}')} e^{j\omega(t - t')} d^3\vec{r} dt.$$

Note that the above definition of the Fourier transform assumes $e^{j[\vec{k} \cdot (\vec{r} - \vec{r}') - j\omega(t - t')]}$ as the basis function. Performing the Fourier transform to the wave equation above, we get:

$$-k^2 g(\vec{k}, \omega) + \frac{\omega^2}{c^2} g(\vec{k}, \omega) = -1,$$

from which we readily obtain:

$$g(\vec{k}, \omega) = \frac{1}{k^2 - \frac{\omega^2}{c^2}}.$$

The Green's function can be calculated by the inverse Fourier transform of $g(\vec{k}, \omega)$:

$$G(\vec{r}, t; \vec{r}', t') = \frac{1}{(2\pi)^4} \iint g(\vec{k}, \omega) e^{i\vec{k} \cdot (\vec{r} - \vec{r}')} e^{-i\omega(t - t')} d\omega d^3\vec{k}$$

You can skip this part until the end of the derivation

$$\begin{aligned} &= \frac{1}{16\pi^4} \iint \frac{1}{k^2 - \frac{\omega^2}{c^2}} e^{i\vec{k} \cdot \vec{r}} e^{-i\omega t} d\omega d^3\vec{k} \\ &= \frac{1}{16\pi^4} \cdot c \iint \frac{e^{j\vec{k} \cdot \vec{r}} e^{-j\omega t}}{k^2 - \frac{\omega^2}{c^2}} d\left(\frac{\omega}{c}\right) d^3\vec{k} \\ &= \frac{c}{16\pi^4} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \frac{e^{-j\frac{\omega}{c} t} d\left(\frac{\omega}{c}\right)}{\left(k + \frac{\omega}{c}\right) \left(k - \frac{\omega}{c}\right)} \right] e^{j\vec{k} \cdot \vec{r}} d^3\vec{k} \\ &= \frac{c}{16\pi^4} \int e^{j\vec{k} \cdot \vec{r}} \left[\frac{1}{2k} \int_{-\infty}^{\infty} \frac{e^{-jx\tau c}}{k + x} dx + \frac{1}{2k} \int_{-\infty}^{\infty} \frac{e^{-jx\tau c}}{k - x} dx \right] d^3\vec{k} \\ &= \frac{c}{16\pi^4} \int e^{j\vec{k} \cdot \vec{r}} \left[\frac{1}{2k} \left(\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{-jx\tau c}}{x + k} dx + \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{-jx\tau c}}{k - x} dx \right) \right] d^3\vec{k} \\ &\quad (\text{P.V. denotes principal value}) \\ &= -\frac{c}{16\pi^4} \int e^{j\vec{k} \cdot \vec{r}} \left[\frac{1}{2k} (j\pi e^{jk\tau c} - j\pi e^{-jk\tau c}) \right] d^3\vec{k} \\ &= -\frac{c \cdot j}{8\pi^3} \int e^{j\vec{k} \cdot \vec{r}} \left(\frac{e^{jk\tau c} - e^{-jk\tau c}}{2k} \right) d^3\vec{k} = -\frac{c \cdot j}{8\pi^3} \int e^{j\vec{k} \cdot \vec{r}} \left(\frac{2j \sin(ck\tau)}{2k} \right) d^3\vec{k} \\ &= \frac{c}{4\pi^2} \int_0^{\infty} \int_0^{\pi} k e^{jkz \cos \theta} \sin(ck\tau) \sin \theta d\theta dk \\ &= -\frac{c}{4\pi^2} \int_0^{\infty} \int_0^{\pi} k e^{jkz \cos \theta} \sin(ck\tau) d\cos \theta dk \\ &= -\frac{c}{4\pi^2} \int_0^{\infty} dk k \sin(ck\tau) \int_1^{-1} e^{jkz x} dx \\ &= -\frac{c}{4\pi^2 j z} \int_0^{\infty} dk \sin(ck\tau) \int_1^{-1} e^{jkz x} d(jkz x) \\ &= -\frac{c}{4\pi^2 j z} \int_0^{\infty} \sin(ck\tau) (e^{-jkz} - e^{jkz}) dk \end{aligned}$$

$$\begin{aligned}
&= \frac{2jc}{4\pi^2 j \dot{z}} \int_0^\infty \sin(ck\tau) \sin(k\dot{z}) dk \\
&= \frac{c}{4\pi^2 \dot{z}} \int_0^\infty [\cos(ck\tau - k\dot{z}) - \cos(ck\tau + k\dot{z})] dk
\end{aligned}$$

In the distributional sense, we use $\int_0^\infty \cos(ka) dk = \pi\delta(a)$ to get

$$= \frac{c}{4\pi \dot{z}} [\delta(c\tau - \dot{z}) - \delta(c\tau + \dot{z})]$$

Applying the scaling property of the delta function ($\delta(ax) = \frac{\delta(x)}{a}$), we can write the above delta functions in a more comprehensive way to get

$$= \frac{1}{4\pi \dot{z}} \left[\delta\left(\tau - \frac{\dot{z}}{c}\right) - \delta\left(\tau + \frac{\dot{z}}{c}\right) \right].$$

The term $\frac{\delta(\tau - \frac{\dot{z}}{c})}{\dot{z}}$ represents a diverging spherical wave, while $\frac{\delta(\tau + \frac{\dot{z}}{c})}{\dot{z}}$ corresponds to a converging spherical wave. Causality requires us to retain only the diverging wave.

End of derivation

Finally, we reach the expression of the Green's function:

$$G(\vec{r}, t; \vec{r}', t') = \frac{1}{4\pi} \frac{\delta\left(t - t' - \frac{\dot{z}}{c}\right)}{\dot{z}}$$

We can use the above Green's function to construct the result:

$$\phi(\vec{r}, t) = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{r}', t') \frac{\delta\left(t - t' - \frac{|\vec{r} - \vec{r}'|}{c}\right)}{|\vec{r} - \vec{r}'|} dt' d^3\vec{r}'.$$

By employing the sifting property of the delta function, we readily obtain:

$$\phi(\vec{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho\left(\vec{r}', t - \frac{\dot{z}}{c}\right)}{\dot{z}} dV'$$

The same principle applied to get the expression for the vector potential:

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}\left(\vec{r}', t - \frac{\dot{z}}{c}\right)}{\dot{z}} dV'.$$

These expressions are similar to the formulas for electric and magnetic potentials we encountered in electrostatics and magnetostatics. However, in this case, the potential at the current time t is determined by the charge and current distributions at an earlier time, known as the retarded time $t_r = t - \frac{\dot{z}}{c}$. This adjustment accounts for the finite propagation speed of electromagnetic waves, which travel at the speed of light c . Consequently, the potentials expressed in these formulas are called retarded potentials.

To better understand how the retarded potential formulas are used in practice, consider the following example: dynamic charge and current distributions are confined within a spherical region, as depicted in the figure. To numerically evaluate the electric potential, we divide the source region into thin spherical shells and integrate the contributions from the charge distribution in each shell. The retarded time for each shell's contribution varies based on its distance from the field point, as illustrated in the figure.

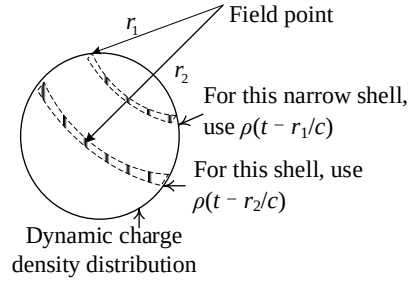


Figure. Illustration of how retarded potential is calculated.

Once the retarded potentials are calculated, we can compute the electric and magnetic fields. However, field calculations cannot directly mirror the retarded potential formulas. For instance, there is no "retarded" equivalent of Coulomb's law, such as:

$$\vec{E}(\vec{r}, t) \neq \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}', t - \frac{r}{c})}{r^2} \hat{r} dV'.$$

To compute the electric and magnetic fields directly from the charge density ρ and current density $\vec{J}$, one can use Jefimenko's equations (see Griffiths' Introduction to Electrodynamics). These equations, while comprehensive, are more complex and computationally intensive than evaluating the retarded potentials.

③ Electric dipole radiation

In this subsection, we present a rigorous analysis of a classic problem: electric dipole radiation. Consider an electric dipole located at the origin, with its dipole moment aligned along the $\hat{z}$ direction. The two charges oscillate harmonically with time, causing both the dipole moment $\vec{d}$ and the associated polarization $\vec{p}$ to vary sinusoidally. These charge oscillations produce a time-varying current, also directed along $\hat{z}$. The oscillating dipole and the corresponding spherical coordinate system are depicted in the accompanying figure.

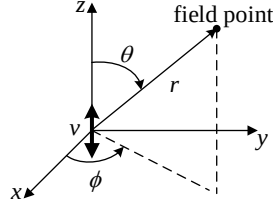


Figure. Oscillating dipole in a spherical coordinate system.

To evaluate the vector potential $\vec{A}$ we proceed with a series of reasonable approximations. The retarded potential formula is given by:

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}', t - r/c)}{r} dV'$$

If the dipole is small compared to r , we can simplify the integral. Specifically, $r \approx r'$, and $t - r/c \approx t - r'/c$, provided the oscillation frequency is sufficiently low (i.e., the oscillation period is much larger than the light travel time across the dipole). Moreover, note that $\int \vec{J} dV = \sum_{i=1}^2 q_i \vec{v}_i$ (in the case of discrete charge distribution, see Eq. 5.30 of Griffith's book), the potential can be expressed as

$$\begin{aligned} \vec{A}(\vec{r}, t) &= \frac{\mu_0}{4\pi r} \int \vec{J}(\vec{r}', t - r/c) dV' \\ &= \frac{\mu_0}{4\pi r} \sum q_i \vec{v}_i(t - r/c) \\ &= \frac{\mu_0}{4\pi r} \frac{d}{dt} \underbrace{\sum q_i \vec{r}_i(t - r/c)}_{\text{total dipole moment at } t_r} \\ &= \frac{\mu_0}{4\pi r} \frac{\partial}{\partial t} \vec{p}(t - r/c) \\ &= \frac{\mu_0}{4\pi r} \dot{\vec{p}}(t - r/c) \hat{z}. \end{aligned}$$

To compute the magnetic field, it is advantageous to use a spherical coordinate system, as it leverages the symmetry of the problem. In spherical coordinates, $\vec{B} = \nabla \times \vec{A}$ can be evaluated as:

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta A_\phi) - \partial_\phi A_\theta] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \partial_\phi A_r - \partial_r (r A_\phi) \right] \hat{\theta} + \frac{1}{r} [\partial_r (r A_\theta) - \partial_\theta A_r] \hat{\phi}.$$

Since in this particular case, $\vec{A}$ is aligned along the $\hat{z}$ direction, it can be expressed in terms of its components in the spherical coordinate system, specifically along the $\hat{\theta}$ and $\hat{r}$ directions (i.e., $A_r = A \cos \theta$, $A_\theta = -A \sin \theta$), as illustrated in the figure below. Moreover, our analysis above suggests that $\vec{A}$ is only dependent on r , but not on θ and ϕ .

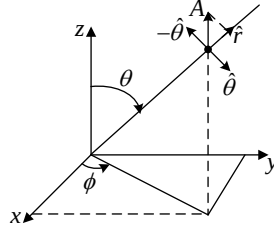


Figure. Decomposing $\vec{A}$ along the $\hat{\theta}$ and $\hat{r}$ directions.

With these conditions, most terms in $\nabla \times \vec{A}$ vanishes, the equation is significantly reduced to:

$$\nabla \times \vec{A} = \frac{1}{r} [\partial_r(rA_\theta) - \partial_\theta A_r] \hat{\phi}$$

in which,

$$\begin{cases} A_\theta = -\frac{\mu_0}{4\pi r} \dot{p}(t - r/c) \sin\theta \\ A_r = \frac{\mu_0}{4\pi r} \dot{p}(t - r/c) \cos\theta \end{cases}$$

As a result,

$$\vec{B} = -\frac{\mu_0}{4\pi r} \sin\theta \left[\partial_r \dot{p}(t - r/c) - \frac{1}{r} \dot{p}(t - r/c) \right] \hat{\phi}.$$

Set $u = t - r/c$, so $\partial_r \dot{p}(t - r/c) = \frac{d\dot{p}}{du} \cdot \frac{du}{dr} = -\frac{1}{c} \ddot{p}(t - r/c)$, and we get:

$$\begin{aligned} \vec{B} &= \frac{\mu_0}{4\pi r} \left[\frac{1}{r} \dot{p}(t - r/c) + \frac{1}{c} \ddot{p}(t - r/c) \right] \sin\theta \hat{\phi} \\ &= \frac{\mu_0}{4\pi c} \left[\frac{c}{r^2} \dot{p}(t - r/c) + \frac{1}{r} \ddot{p}(t - r/c) \right] \sin\theta \hat{\phi}. \end{aligned}$$

This is the magnetic field $\vec{B}$ produced by an oscillating electric dipole. The second term, proportional to the dipole moment's second time derivative (i.e., $\ddot{p}$), represents the radiation field, which decays as $1/r$. Additionally, the radiation field magnitude varies with angle as $\sin\theta$. These characteristics are consistent with the findings from our earlier analysis using a simple accelerating charge model. Given that the dipole undergoes harmonic oscillatory motion, we can represent the relevant quantities using complex exponentials:

$$\begin{aligned} \tilde{\vec{p}} &= \tilde{p}_0 e^{-j\omega t} \hat{z}, \quad \tilde{p}(t - r/c) = \tilde{p}_0 e^{-j[\omega(t - r/c)]} = \tilde{p}_0 e^{j(kr - \omega t)}, \\ \dot{\tilde{p}}(t - r/c) &= -j\omega \tilde{p}_0 e^{j(kr - \omega t)}, \\ \ddot{\tilde{p}}(t - r/c) &= -\omega^2 \tilde{p}_0 e^{j(kr - \omega t)}, \end{aligned}$$

Substituting the above expressions into the formula for the magnetic field, we obtain:

$$\tilde{\vec{B}}(\vec{r}, t) = -\frac{\mu_0 \omega \tilde{p}_0}{4\pi c} \left[j \frac{c}{r^2} + \frac{\omega}{r} \right] \sin\theta e^{j(kr - \omega t)} \hat{\phi}$$

$$= -\frac{\tilde{p}_0 k^3}{4\pi\epsilon_0 c} \left[\frac{j}{(kr)^2} + \frac{1}{(kr)} \right] \sin\theta e^{j(kr-\omega t)} \hat{\phi}.$$

Similarly, the electric field can be calculated using the equation $\vec{E} = -\nabla\phi - \partial_t \vec{A}$, where the expression for the scalar potential can be derived from the Lorenz gauge condition:

$$\partial_t \phi = -c^2 \nabla \cdot \vec{A}.$$

The electric field is found to exhibit both r and θ components:

$$\begin{aligned} \tilde{E}_r &= \frac{2\tilde{p}_0 k^3}{4\pi\epsilon_0} \left[\frac{1}{(kr)^3} - \frac{j}{(kr)^2} \right] \cos\theta e^{j(kr-\omega t)} \\ \tilde{E}_\theta &= \frac{\tilde{p}_0 k^3}{4\pi\epsilon_0} \left[\frac{1}{(kr)^3} - \frac{j}{(kr)^2} - \frac{1}{kr} \right] \sin\theta e^{j(kr-\omega t)} \end{aligned}$$

When $kr < 1$, the term associated with $\frac{1}{(kr)^3}$ dominates, which is referred to as the near field.

Recall that a static electric dipole has a field pattern given by:

$$\vec{E}_{\text{dipole}}(\vec{r}) = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta}).$$

In the near field, the field pattern mirrors that of a static dipole, except for the factor $e^{j(kr-\omega t)}$, which accounts for wave propagation.

In the far field, where $kr \gg 1$, the term associated with $\frac{1}{kr}$ dominates. Interestingly, only the θ component of the electric field contains this radiation component, and it also exhibits an angular dependence of $\sin\theta$, similar to the radiation component of the magnetic field.

Suppose a radiator emits a fixed amount of power. On the sphere of radius $|\vec{r}|$ centered on the radiator, the total energy should integrate to a constant value. The area of the sphere is proportional to r^2 , and the power density is $|\vec{E} \times \vec{B}| \propto \frac{1}{r^2}$. Therefore, both $E(r)$ and $B(r)$ must decay as $1/r$, justifying why the radiation component should decay as $1/r$.

④ The complex Poynting vector

When the field is monochromatic (i.e., of a single frequency), we use complex notations, as is done when dealing with the dipole radiation problem. It is essential to expand on this concept to encompass the notion of energy flow, for which we employ the Poynting vector. In the context of a complex field, we develop the complex Poynting vector as follows. First, we link the true fields with their complex counterparts:

$$\begin{aligned} \vec{E}(\vec{r}, t) &= \text{Re} \left\{ \tilde{\vec{E}}(\vec{r}) e^{-j\omega t} \right\} \\ &= \frac{1}{2} \left\{ \tilde{\vec{E}}(\vec{r}) e^{-j\omega t} + \tilde{\vec{E}}^*(\vec{r}) e^{j\omega t} \right\}, \end{aligned}$$

$$\vec{B}(\vec{r}, t) = \frac{1}{2} \{ \tilde{\vec{B}}(\vec{r}) e^{-j\omega t} + \tilde{\vec{B}}^*(\vec{r}) e^{j\omega t} \}.$$

The Poynting vector is defined as

$$\begin{aligned} S(\vec{r}, t) &= \frac{1}{\mu_0} \vec{E}(\vec{r}, t) \times \vec{B}(\vec{r}, t) \\ &= \frac{1}{\mu_0} \left\{ \frac{1}{2} (\tilde{\vec{E}}(\vec{r}) e^{-j\omega t} + \tilde{\vec{E}}^*(\vec{r}) e^{j\omega t}) \times \frac{1}{2} (\tilde{\vec{B}}(\vec{r}) e^{-j\omega t} + \tilde{\vec{B}}^*(\vec{r}) e^{j\omega t}) \right\} \\ &= \frac{1}{4\mu_0} \left\{ \tilde{\vec{E}} \times \tilde{\vec{B}}^* + \tilde{\vec{E}}^* \times \tilde{\vec{B}} + \tilde{\vec{E}} \times \tilde{\vec{B}} e^{-j2\omega t} + \tilde{\vec{E}}^* \times \tilde{\vec{B}}^* e^{j2\omega t} \right\} \\ &= \frac{1}{4\mu_0} \left\{ \tilde{\vec{E}} \times \tilde{\vec{B}}^* + (\tilde{\vec{E}} \times \tilde{\vec{B}}^*)^* + \tilde{\vec{E}} \times \tilde{\vec{B}} e^{-j2\omega t} + (\tilde{\vec{E}} \times \tilde{\vec{B}} e^{-j2\omega t})^* \right\} \\ &= \frac{1}{2\mu_0} \text{Re} \left\{ \tilde{\vec{E}} \times \tilde{\vec{B}}^* \right\} + \frac{1}{2\mu_0} \left\{ \tilde{\vec{E}} \times \tilde{\vec{B}} e^{-j2\omega t} \right\}. \end{aligned}$$

The instantaneous power density consists of two components: the first component is time-independent, while the second oscillates at a very high frequency of 2ω . When considering the average intensity, the second term is effectively smoothed out, leaving only the first term. As a result, we obtain (we use the symbol “ $\langle \rangle$ ” for time averaging):

$$\begin{aligned} \langle \vec{S}(\vec{r}) \rangle &= \frac{1}{2} \text{Re} \left\{ \tilde{\vec{E}} \times \frac{\tilde{\vec{B}}^*}{\mu_0} \right\} = \text{Re} \left\{ \frac{1}{2} \tilde{\vec{E}} \times \tilde{\vec{H}}^* \right\} \\ &= \text{Re} \left\{ \tilde{\vec{S}} \right\} \end{aligned}$$

where $\tilde{\vec{S}} = \frac{1}{2} \tilde{\vec{E}} \times \tilde{\vec{H}}^*$ is the complex Poynting vector.

Example 1: plane waves

In free space, a plane wave obeys $j\vec{k} \times \tilde{\vec{E}} = j\omega\tilde{\vec{B}}$ (it is a direct consequence of $\nabla \times \vec{E} = -\partial_t \vec{B}$

and the wave factor $e^{j(\vec{k} \cdot \vec{r} - \omega t)}$ in $\tilde{\vec{E}}$ and $\tilde{\vec{B}}$)

$$\tilde{\vec{B}} = \left(\frac{k}{\omega} \right) \hat{k} \times \tilde{\vec{E}} = \frac{1}{c} \hat{k} \times \tilde{\vec{E}}.$$

Consequently, $\vec{E} \times \vec{B}$ is along $\hat{k}$, and $\tilde{\vec{E}}$ and $\tilde{\vec{B}}$ share the same phase.

$$\text{Re} \left\{ \tilde{\vec{S}} \right\} = \text{Re} \left\{ \frac{1}{2\mu_0} \tilde{\vec{E}} \times \tilde{\vec{B}}^* \right\} = \text{Re} \left\{ \frac{1}{2\mu_0} \tilde{\vec{E}} \times \left(\frac{k}{\omega} \right) \hat{k} \times \tilde{\vec{E}}^* \right\}$$

For linearly polarized wave, $\tilde{\vec{E}}$ can be expressed as $\tilde{\vec{E}} = \tilde{E} \hat{u} = E e^{j\varphi} \hat{u}$ where $\hat{u}$ is the unit vector denoting the direction of the electric field,

$$\text{Re} \left\{ \tilde{\vec{S}} \right\} = \frac{1}{2\mu_0 c} E^2 \hat{k} \propto \tilde{\vec{E}} \cdot \tilde{\vec{E}}^*$$

That is why in many textbooks, people use $\tilde{\vec{E}} \cdot \tilde{\vec{E}}^*$ to calculate the light intensity.

Example 2: Dipole radiation

In the near field ($rk \ll 1$), the $\frac{1}{(kr)^3}$ term dominates the electric field, while the $\frac{1}{(kr)^2}$ term dominates the magnetic field. However, the near fields of E and B has a phase difference of $\frac{\pi}{2}$, thus

$\tilde{\vec{E}} \times \tilde{\vec{B}}^*$ (near field component) is purely imaginary, yielding $Re\{\tilde{\vec{S}}\} = 0$.

When $rk \gg 1$, the $\frac{1}{(rk)}$ terms dominate, so in the far field,

$$\begin{cases} \tilde{\vec{E}}(\vec{r}) \approx -\frac{\tilde{p}_0 k^2}{4\pi\epsilon_0} \cdot \frac{1}{r} \sin\theta e^{jkr} \hat{\theta} \\ \tilde{\vec{B}}(\vec{r}) \approx -\frac{\tilde{p}_0 k^2}{4\pi\epsilon_0} \cdot \frac{1}{r} \sin\theta e^{jkr} \hat{\phi} \end{cases}$$

The situation is shown in the following figure.

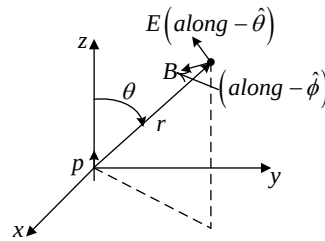


Figure. Directions of E and B fields in the far field of a dipole radiator.

With the analysis, the following conclusions are reached:

- (a) $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}^*$ is along $\hat{r}$, representative of an outgoing wave;
- (b) The complex Poynting vector can be expressed as

$$\langle \vec{S}(\vec{r}) \rangle = \frac{1}{2} Re \left\{ \tilde{\vec{E}} \times \frac{\tilde{\vec{B}}^*}{\mu_0} \right\} = \frac{\pi^2 p_0^2 f^4 \sin^2 \theta}{2\epsilon_0 c^3 r^2} \hat{r},$$

which decays as $1/r^2$ and exhibits an angular dependence of $\sin^2 \theta$. If $\theta = 30^\circ$, $\sin^2 \theta = 1/4$; if $\theta = 60^\circ$, $\sin^2 \theta = 3/4$. A polar plot of the angular distribution is shown below.

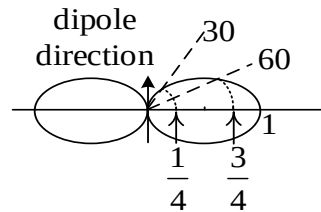


Figure: Polar plot of the angular distribution of the far-field intensity of dipole radiation